# ALAN TSENG

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## EDUCATION

**Master of Computer Science**, National Taiwan University 2021-2024
GPA: 4.23/4.3

## EXPERIENCE

**Software Engineer Intern** Jul 2022 - Nov 2022
Line Corporation (Blockchain, React, Nx, Kubernetes, Swagger) Taipei, Taiwan

- Developed and deployed a Swagger-based API development tool, streamlining API development by 20% and ensuring single source of truth. Seamless Kubernetes integration with optimized performance through DNS and load balancing configuration.
- Leveraged Nx to develop a scalable frontend architecture within a monorepo structure, and collaboration with the Korean backend team.
- Led a team of 5 developers in an internal hackathon to build a merchant management system using NextJS and Spring Boot.
- Presented a talk about Ethereum and Decentralized Application Development within an internal study group.

## PROJECTS

**FansTick**
Master's Thesis (Solidity, TypeScript, React Native, Python, IPFS, GunDB)

- TBD

**Human tremor symptoms detection**
Academic Thesis (Python, Tensorflow)

- Developed an LSTM-RNN model with Tensorflow Functional API for tremor frequency and noise detection.
- Achieved an average accuracy of 94.8% on a dataset of 160,000 simulated signals.

**Vaccine Tracking System**
Course Project (Python, Solidity)

- Developed a blockchain-based vaccine tracking system utilizing Bloom Filter for efficient and rapid vaccine batch number verification with lower time complexity O(k) and space complexity O(m).

**Drum.io, Game**
Course Project (JavaScript, C, SocketIO, Arduino, Tensorflow.js)

- Designed and implemented a KNN-based multiplayer air drum game, awarded 1st place in Interactive Media Programming (52 participants) and 3rd place by collective vote (50 participants).
- Real-time analysis of acceleration data from an embedded device enabled intuitive air drumming and facilitated online competition.

**Create your dancing robot, Computer Graphics**
Course Project (C/C++, OpenGL)

- Got First place in the Computer graphics final project in a collective vote of 52 people.
- Implemented a model control system in OpenGL, enabling a robot to jump, inspired by Maya's graphical model control system.

**Landing Planet, Game**
Course Project (Java, Android)

- Won second place in a space exploration game development competition, with the support of 50 participants.
- Developed a space exploration game using Java and object-oriented concepts.

**Rate My Professors TW, Full stack project**
Side Project (NextJS, MVC, JWT, CI/CD, Web Crawling, NodeJS, MongoDB, PWA)

- Efficiently crawled course information from MCU-up and the school's course page using asyncio and aiohttp, reducing the crawling time by 83%.
- Built a course scheduler and course rating system using NextJS, and implemented a REST API with FastAPI and MongoDB.

## EXTRA-CURRICULAR ACTIVITIES

**Developer Student Club Leader** (2019-2021) - Advisor by Yuh-Pyng, Shieh and Google Developers

- Selected as one of 800 leaders among over 3,000 passionate applicants worldwide for driving peer learning and community in a 2-year professional development program.
- Led and grew a technical student club, fostering a vibrant community and engaging over 73 attendees through talks on Deep Learning, PoseNet, Python & Colab, and Flutter.

**ACM Competitive Programing Club Leader** (2017-2021) - Supervised by Jeng-Sheng Yeh

- Enhanced programming skills through practice and problem-solving on platforms like UVa Online Judge.

**Volunteered at exhibitions**

- Assisted with Developer Student Club (DSC) and Google Developer Groups (GDG) events at SITCON(Student Information Technology Conference) and COSCUP(Conference for Open Source Coders, Users and Promoters).
- Provided technical support at the ART FUTURE exhibition showcasing blockchain applications.

## HONORS

**Research Excellence Award**

- Recognized for outstanding research contributions, placing among the top 1% of graduating students.

**Intel Andy Grove Scholarship**

- Selected as one of 367 recipients worldwide, out of over 2,000 applicants, for demonstrating academic excellence and promising accomplishments.

**College Student Research Scholarship**

- Awarded a prestigious scholarship by the Ministry of Science and Technology, Taiwan.

## PUBLICATIONS

- Sung, H.-M., Chen, T., **Tseng, H.-C.**, Prayogo, B., Lin, J.-Y., & Hung, Y.-P. (2023). **akaTick: Hybrid Mobile E-Ticketing System Based on Non-Fungible Tokens**. 2023 IEEE International Conference on Metaverse Computing, Networking and Applications (MetaCom), 686–687.
- **Tseng, H.-C**., Tu, C.-W., Huan, X.-Y., & Chia, T.-L. (2020). **Nighttime vehicle light detection based on deep learning and image enhancement**. 2020 IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP), 1–8.

## SKILLS

- Blockchain(EVM Compatible Chains(L1/L2), Tezos, COSMOS, Solana), Artificial Intelligence(Machine Learning, DeepLearning), Web(Frontend/Backend) Development, Cloud-Native Applications, Computer Vision, Socket, Chrome Extension, SSG/SSR, MVC,

- Python, C/C++, JavaScript/TypeScript, Solidity, Java, SmartPy, Swift, Dart, C#, Matlab
- FastAPI/Flask, NodeJS, React(NextJS, React Native), Tensorflow, Scikit-Learn, OpenCV, OpenGL, Numpy, Pandas, Matplotlib, Processing(p5), Flutter
- NoSQL(MongoDB), SQL(PostgreSQL, MySQL), p2p(orbitDB, GunJS), IPFS